

GentleWoman Blueprint v1.0 — Solution Blueprint

Solution Blueprint

GentleWoman — Headless Commerce Storefront x Odoo 19

Erajaya — Value-Added Services *Architecture & Integration Blueprint*

Document Control / Kendali Dokumen

Item	Detail
Document Title	Solution Blueprint — GentleWoman Headless Commerce
Document ID	GW-BP-001
Version	1.0
Status	Final for Approval
Classification	Confidential — Internal
Date	Juni 2026
Author / Owner	Product Owner — Value-Added Services (Erajaya)
Audience	Architecture, Engineering, Security, Delivery Leadership
Related Documents	TSD (GW-TSD-001), FSD (GW-FSD-001), Business Presentation (GW-PRES-001)

Revision History / Riwayat Revisi

Version	Date	Author	Description
1.0	Juni 2026	Product Owner — VAS	Initial release / Rilis awal

Approval / Lembar Pengesahan

Role	Name	Signature	Date
Prepared by — Product Owner, Value-Added Services			
Reviewed by — IT Business Analyst / Technical Lead			
Approved by — Head of Value-Added Services / IT			

Catatan bahasa / Language note. Bilingual: Bahasa Indonesia + English technical terms.

1. Konteks Eksekutif / Executive Context

ID. GentleWoman adalah inisiatif *headless commerce* untuk fashion: storefront Next.js memberi kendali penuh atas pengalaman brand (editorial, 3D, AI), sementara Odoo 19 menjadi sumber kebenaran tunggal untuk katalog, harga, stok, order, pelanggan, promosi, afiliasi, dan konten. Pendekatan ini memisahkan *presentasi* (frontend) dari *proses bisnis* (backend) sehingga keduanya berevolusi independen tanpa mengorbankan integritas data.

EN. GentleWoman is a *headless commerce* initiative for fashion: the Next.js storefront owns the brand experience (editorial, 3D, AI) while Odoo 19 is the single source of truth for catalog, price, stock, orders, customers, promotions, affiliate, and content. This decouples *presentation* (frontend) from *business processes* (backend), letting each evolve independently without sacrificing data integrity.

2. Solution Blueprint (Logical)

EXPERIENCE LAYER – Storefront (Next.js 15)	
Editorial UI · 3D PDP · AI Shopper · i18n ID/EN · SEO/OG	
INTEGRATION LAYER – BFF (Next.js route handlers)	
/api/store · /api/img · /api/shopper · /api/docs · /api/affiliate	
inject auth + tenant · seal/refresh token · rate-limit	
TLS sidecar	(storefront-tls, cert-pinned)
COMMERCE LAYER – Odoo 19 CE (System of Record)	
custom_storefront_api · custom_affiliate · custom_ecommerce	
custom_payment_id · custom_core · custom_pdp_* · website_sale	
AI LAYER – ai-gateway (FastAPI, multi-provider)	
Claude / Ollama · retrieval · HMAC · rate-limit · prompt cache	
DATA LAYER – PostgreSQL (db-per-tenant: gentlewoman) · Redis	

3. Component Blueprint

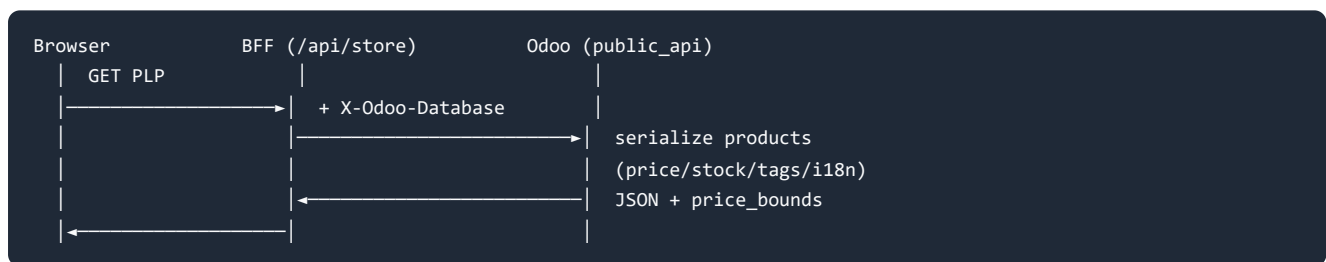
Component	Peran / Role
Storefront (Next.js)	UI + Backend-for-Frontend; pemilik presentasi
BFF route handlers	Batas keamanan; proxy ke Odoo & ai-gateway; auth/tenant injection
storefront-tls	TLS sidecar cert-pinned untuk hop internal BFF↔Odoo
custom_storefront_api	API headless 3-lapis (public/JWT/HMAC), cart/checkout, content, store locator
custom_affiliate	Program afiliasi: link, klik, konversi, payout, atribusi
custom_core	HMAC secure-endpoint, Fernet field encryption
custom_ecommerce / custom_payment_id	Kurir & ongkir; adapter payment (Eraspaces stub)
custom_pdp_*	Klasifikasi data, consent, audit, masking, retention (UU PDP)
ai-gateway	Multi-provider AI (Claude/Ollama), retrieval shopper
PostgreSQL / Redis	Persistensi per-tenant; cache/rate-limit

4. Integration Blueprint

ID. Inti integrasi adalah BFF: browser tidak pernah memegang secret atau menyentuh Odoo langsung. Tiga lapis otentikasi dipakai sesuai sensitivitas.

Layer	Mechanism	Scope
Public	auth=public + CORS	Katalog, PDP, tags, stores, content, shipping quote (read-only)
Customer	JWT (HS256) bearer, sealed HttpOnly cookie	Cart, order, address, wishlist (partner-scoped)
Admin	HMAC-SHA256 (X-Signature / X-Timestamp)	Webhook server-to-server (health, sync, status)

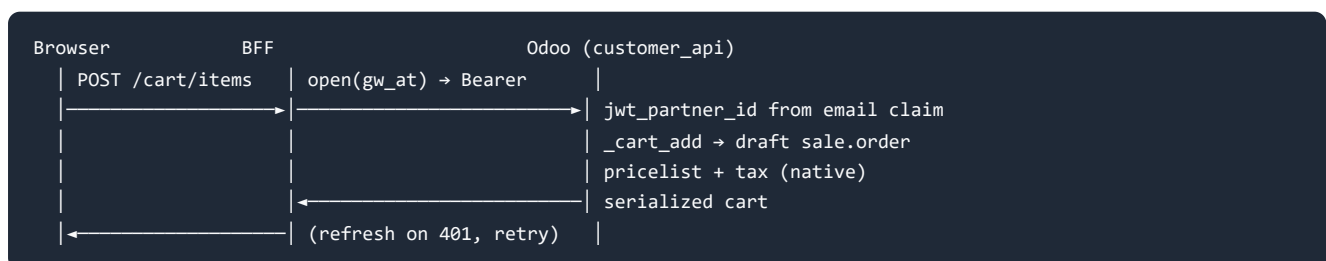
4.1 Sequence — Browse (Public)



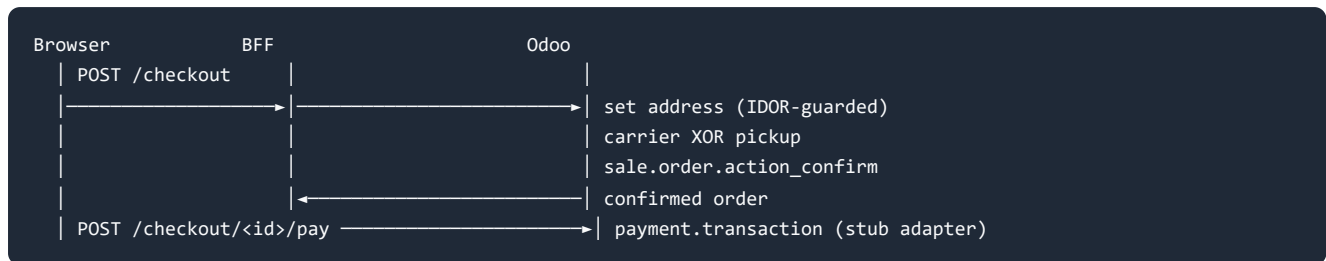
4.2 Sequence — Login & Token



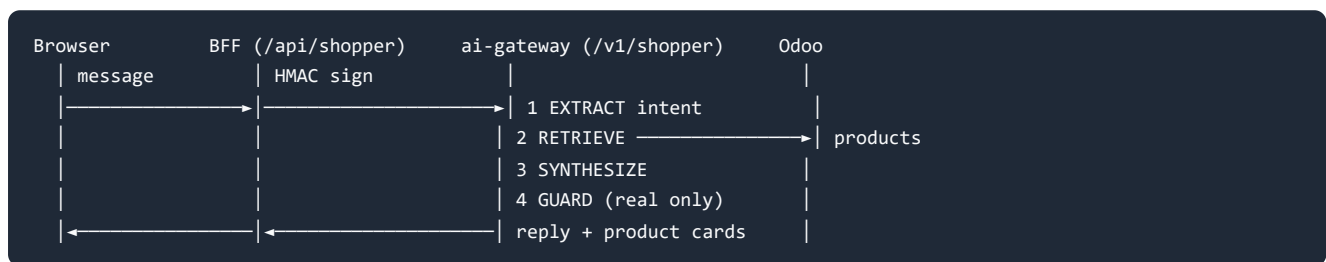
4.3 Sequence — Add to Cart (JWT)



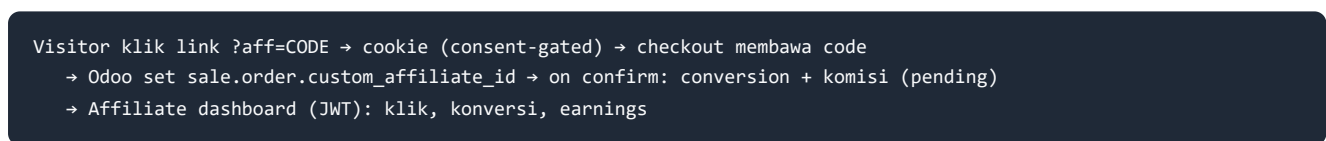
4.4 Sequence — Checkout



4.5 Sequence — AI Personal Shopper



4.6 Sequence — Affiliate Attribution



5. Data Blueprint — Ownership

ID. Odoo adalah pemilik data otoritatif. Storefront tidak menyimpan salinan data bisnis selain state UI sementara.

Entity	Owner	Catatan
Product, variant, tag, category	Odoo	<code>product.*</code> native + ekstensi <code>custom_*</code>
Price & promo	Odoo	<code>pricelist</code> ; <code>compare_at</code> / <code>discount_pct</code> dihitung saat serialize
Stock & in-store availability	Odoo	<code>qty_available</code> ; per-warehouse <code>location</code> -scoped
Cart & order	Odoo	draft → confirmed <code>sale.order</code>
Customer & address	Odoo	<code>res.partner</code> ; PII at-rest Fernet
Affiliate	Odoo	<code>custom.affiliate*</code>
Editorial content & newsletter	Odoo	<code>custom.storefront.content/subscriber</code>
UI state (cart badge, locale)	Storefront	sementara (Zustand)
Session token	Storefront cookie	JWT sealed AES-256-GCM (HttpOnly)

6. Security Blueprint (Defense-in-Depth)

```
[ Transport ]  HTTPS (Caddy) + internal TLS sidecar (cert-pinned)
[ Session ]   HttpOnly cookie + AES-256-GCM sealed + transparent refresh + hashed refresh
[ App ]       strict CSP · DOMPurify · anti-SSRF image proxy · rate-limit + body cap · anti-IDOR
[ Data ]      PII Fernet at-rest · ORM-only (no raw SQL) · audit (custom_pdp_audit)
[ Privacy ]   UU PDP consent capture · retention · DSAR/masking (custom_pdp_*)
[ Tenancy ]   DB-per-tenant · X-Odoo-Database · HMAC tenant allow-list
[ Secrets ]   externalized env/secret config · rotation procedures
```

7. Deployment Blueprint

```
Linux Host (Docker Engine + Compose)
├── Caddy (TLS LB)
│   ├── storefront (Next.js, read-only FS, non-root)
│   │   └── storefront-tls → odoo:8069
│   ├── odoo (workers)
│   ├── ai-gateway (FastAPI)
│   └── nginx (shared)
├── postgres (db-per-tenant: gentlewoman)
├── redis (cache/RL)
└── Persistent volumes: filestore · DB · logs · backup
```

- **Isolasi tenant.** Perubahan storefront ter-scope ke `gentlewoman` ; tenant lain tidak terpengaruh.
- **Deploy.** Storefront = rebuild image + recreate; modul Odoo = `-u <module> -d gentlewoman` .

8. AI Blueprint

ai-gateway (FastAPI)

- └ provider abstraction: Claude (Haiku/Sonnet, prompt cache) | Ollama (qwen2.5 / llama3.2)
- └ /v1/shopper: EXTRACT → RETRIEVE(Odoo) → SYNTHESIZE → GUARD
- └ HMAC-validated calls · per-IP rate-limit · Redis cache
- └ privacy: consent-gated visitor profile, limited retention

9. Environment & Infrastructure

- **OS/Runtime.** Linux (Docker 24+ / Compose v2); container non-root; image hardened.
- **Network.** Bridge internal; hanya Caddy expose 80/443; TLS ACME auto-renew.
- **Observability.** Prometheus/Grafana/Loki; health check storefront & Odoo.
- **Backup.** pg_dump per-tenant + filestore; restore terdokumentasi.

10. Non-Functional & SLA

Aspek	Target
Availability	99.5% (mengikuti SLA platform)
Security posture	Defense-in-depth (§6) terpenuhi
Privacy	UU PDP compliant
Scalability	Storefront stateless (scale-out); rate-limit → Redis saat multi-instance
i18n	ID & EN penuh

11. Roadmap / Phasing

Fase	Cakupan
Fase 1 (Live Pilot)	Catalog/PLP/PDP, 3D PDP, AI Shopper MVP, Cart/Checkout/Wishlist, Store Locator + Click&Collect, Affiliate, i18n, Promo, Security/PDP
Fase 2	CMS lanjutan, share-to-social lengkap, AI Admin (auto-deskripsi/tagging)
Fase 3	Avatar / Virtual Try-On (2D→3D/AR), integrasi payment Eraspaces (saat API tersedia)

Appendix — Glossary

Term	Arti
BFF	Backend-for-Frontend (route handler Next.js sisi server)
PLP / PDP	Product List Page / Product Detail Page
HMAC	Hash-based Message Authentication Code (integritas pesan server-to-server)
JWT	JSON Web Token (otentikasi customer)
IDOR	Insecure Direct Object Reference (dicegah oleh ownership check)
SSRF	Server-Side Request Forgery (dicegah oleh image proxy terkunci)
UU PDP	Undang-Undang Pelindungan Data Pribadi (UU 27/2022)
SoR	System of Record (Odoo sebagai sumber kebenaran)